



Integrated Himalayan EWS – System Architecture

Binding monsoon flooding & landslides / ice–rock avalanche / GLOF / major earthquake onto the two systems Nepal already runs (working draft)

Drafted 2026-08-30 · Revised 2026-09-03 · Based on analysis of the 26 August 2026 Bhote Koshi–Trishuli flood · Not an official position

Do not build from zero. Connect the two systems that exist; add only the missing detection paths.

1 ☀ The two systems Nepal already has

Rain	DHM rainfall & river “live watch”: 280 stations
	Web telemetry since 2010. The backbone of the rainfall track
Rain	Threshold-based flood & landslide warning (since 2018)
	272 hazard areas delineated; SMS is sent when upstream rainfall crosses threshold
Rain	CBFEWS in 8 river systems · 3-day rainfall forecast
	Koshi, Karnali, West Rapti and others. Narayani is included — already present in this corridor
Rain	1 of 3 planned weather radars
	Installed in the west in 2019. Central and east: not yet
Seis	~40 broadband seismometers · ~36 accelerometers
	DMG / NEMRC, plus 8 Univ. of Tokyo–JICA stations and 10 China CEA stations

For rainfall-driven hazards, the chain from detection to dissemination is already complete. What failed here was not the absence of a system but the fact that **the system presupposes rain**.

2 🇳🇵 So why did nothing fire?

Gap 1	The rainfall track presupposes rainfall
	No rain was involved in this event (confirmed by DHM). Neither the 280 stations nor the 272 area thresholds could respond in principle
Gap 2	The seismic track presupposes earthquakes
	Associators are built on sharp arrivals. A slow-onset mass movement is discarded at association. nst = 0 in this event’s catalogue entries is the evidence
Gap 3	The two tracks are not wired together
	Even with an M5.2 on the record (GFZ: Mw 5.7), nothing carries it into the flood alerting path. Both 08:37 and 11:45 reached the catalogue only after an analyst processed them — posted a day and a half later
Gap 4	Mandates are split
	Earthquakes = DMG / flood & landslide warning = DHM / response = NDRRMA

Each track works for its own hazard. Neither owned this hazard, and they were not connected to each other. That is what “integration” has to fix — and none of Gaps 1–4 is a shortage of hardware.

3 ⚡ Hazard stocktake — what is caught, and by which path

HAZARD	DETECTION PATH	STATUS
Monsoon flooding & landslides Cumulatively the largest killer of all	R rainfall threshold + soil water index	existing operating
Ice–rock avalanche / rock collapse This event = Langtang Lirung N face, bedrock and all / 2015 Langtang	D1 single-force source alert in 60–120 s	new
GLOF / landslide-dam breach This event’s initial mechanism + the barrier lake of 28 Aug	D2 fluvial tremor (1–10 Hz)	new
Loss of the instruments themselves Upstream gauges went silent together after 08:40; the 30 Aug rise was noticed by ear	D3 missing data = alarm	new
Widespread coseismic slope failure 25,000 landslides in Gorkha 2015	State modifier lowers every trigger threshold	new

What this system cannot catch — stated explicitly

Snow avalanche Detectable as a small-magnitude D1 class. Forecasting is impossible — no snowpack observation exists

Earthquake early warning itself Nepal has no national EEW (demonstration stage only)

Air blast Occurred at Langtang 2015 and Chamoli 2021. Damage reaches beyond the flow path

4 🔄 Six-layer architecture — what is reused, what is modified, what is added

